



Quantitative measurement of displacement and strain fields from HREM micrographs

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Abstract

A method for measuring and mapping displacement fields and strain fields from high-resolution electron microscope (HREM) images has been developed. The method is based upon centring a small aperture around a strong reflection in the Fourier transform of an HREM lattice image and performing an inverse Fourier transform. The phase component of the resulting complex image is shown to give information about local displacements of atomic planes and the two-dimensional displacement field can be derived by applying the method to two non-colinear Fourier components. Local strain components can be found by analysing the derivative of the displacement field. The details of the technique are outlined and applied to an experimental HREM image of a domain wall in ferroelectric-ferroelastic PbTiO₃. © 1998 Elsevier Science B.V. All rights reserved.

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1. Introduction

In general, high-resolution electron microscope (HREM) image contrast is not a simple function of the crystal structure and to be interpreted, the experimental results must be compared with image simulations. However, in many cases, the image contrast will be peaked at the atomic column positions which appear either dark or bright depending

on crystal thickness and defocus. Techniques based upon locating the peaks in image contrast and associating these with atomic columns have been used by many authors in order to analyse local variations in structural parameters, such as lattice parameters in epitaxially strained layers [1,2] and compound semiconductors [3,4]. These “peak-finding” methods rely on fitting a two-dimensional reference lattice to a subset of peaks associated with a non-distorted region of the image and measuring local deviations from lattice positions. While these methods work well in many situations, particularly

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for monoatomic structures, they are difficult to apply to images where the image contrast is not represented well by Gaussian-shaped intensity peaks. To overcome this particular problem, the unit cells in the image can be located by cross-correlation [5,6]. These positions are then used to characterise the local deformation of the crystal [7,8].

The technique presented in this paper is somewhat different and concerns the analysis of how the spatial frequency components describing the HREM image vary across the field of view. An HREM lattice image is characterised by the presence of strong Bragg-reflections in its Fourier transform. These strong frequency components are related to the two-dimensional unit cell describing the projected crystalline structure. Choosing two non-colinear reciprocal lattice vectors (Bragg reflections) in the Fourier transform implicitly defines a 2-D lattice in the HREM image to which all variations can be referred. A perfect crystal lattice gives rise to sharply peaked frequency components, with variations in lattice spacings giving rise to diffuse intensities in the Fourier transform centred around the frequencies corresponding to the mean lattice spacings. By forming an image from a strong lattice reflection and the accompanying diffuse frequencies, it is possible to determine local variations in the structure being imaged. Unlike the related technique of creating optical moirés [9], which is successful in revealing lattice distortions, the technique described here allows maps of the local distortion to be measured from the image.

It is well known that the positions of fringes in the image do not necessarily correspond to the atomic plane positions. This is due both to the form of the electron wavefunction emerging from the crystal and to the effect of the objective lens. For example, the position of certain fringes in non-centrosymmetric structures will depend on the specimen thickness, and in both axial [10] and non-axial illumination conditions [11] the objective lens influences the spacing of lattice fringes. For the analysis presented here, however, we only concern ourselves with the relative positions of lattice fringes in the image. The absolute value of lattice parameters will not be measured, only the variations in lattice spacing. The problems associated

with such measurements concern, for example, surface relaxation effects in thin crystals [12], mixed surface layers introduced during sample preparation [13] and misalignments coupled with changes in thickness. These points are discussed in more detail elsewhere [14].

In regions where the structure varies rapidly, for example at an interface, the effect of the objective lens becomes an important factor. This is because the information concerning variations is distributed around the Bragg beams. For a particular frequency, the objective lens can introduce shifts in lattice fringes and changes in contrast (including inversions in contrast). Therefore, if the transfer of the lens is not uniform in the region surrounding the Bragg beam, different parts of the information will be imaged differently. The resulting distortion of the displacement field can be seen, for example, in images of strained multilayers [2,17]. The width of the zone of uniform contrast defines the length scales over which the variations in the lattice fringe positions will be imaged faithfully. In modern microscopes where the low spherical aberration allows uniform transfer for the Bragg beams typically of interest, measurements of displacement fields at medium resolution (above 1 nm) are likely to be reliable. More work is however necessary in order to determine the limiting lateral resolution at which such measurements remain valid.

The theoretical basis of this work was first described in a study of strained metal multilayers [15] and of nanomaterials [16], with a fuller development given more recently [14,17]. A similar treatment has also been proposed by Takeda and Suzuki [18] using the method of analysis first developed in the field of optical interferometry [19]. The theory will be extended to determine the two-dimensional displacement field and the corresponding strain fields.

2. Geometric phase images

An image of a perfect crystal can be described as a Fourier series:

$$I(\mathbf{r}) = \sum_{\mathbf{g}} H_{\mathbf{g}} \exp\{2\pi i \mathbf{g} \cdot \mathbf{r}\}, \quad (1)$$

where $I(\mathbf{r})$ is the intensity in the image at position $\mathbf{r}$, and $\mathbf{g}$ are the periodicities corresponding to the Bragg reflections. The Fourier coefficients $H_{\mathbf{g}}$ are, in general, complex and can be written:

$$H_{\mathbf{g}} = A_{\mathbf{g}} \exp\{iP_{\mathbf{g}}\}, \quad (2)$$

where the modulus $A_{\mathbf{g}}$ gives the amplitude of the set of sinusoidal lattice fringes $\mathbf{g}$, and the phase $P_{\mathbf{g}}$ gives the lateral position of the fringes within the original image.

To describe variations in the image of contrast and fringe position, we allow the Fourier components, $H_{\mathbf{g}}$, to become a function of position [17]:

$$I(\mathbf{r}) = \sum_{\mathbf{g}} H_{\mathbf{g}}(\mathbf{r}) \exp\{2\pi i \mathbf{g} \cdot \mathbf{r}\}. \quad (3)$$

The coefficients $H_{\mathbf{g}}(\mathbf{r})$ can be interpreted as the local value of the Fourier components $H_{\mathbf{g}}$ in the image. To show how these functions can be expressed in Fourier space, we begin with the Fourier transform of the image intensity, $\tilde{I}(\mathbf{k})$, as defined by the following equation:

$$I(\mathbf{r}) = \iint \tilde{I}(\mathbf{k}) \exp\{2\pi i \mathbf{k} \cdot \mathbf{r}\} d\mathbf{k}. \quad (4)$$

The Fourier transform of Eq. (3) is then given by:

$$\tilde{I}(\mathbf{k}) = \sum_{\mathbf{g}} \tilde{H}_{\mathbf{g}}(\mathbf{k}) \otimes \delta(\mathbf{k} - \mathbf{g}) \quad (5)$$

which is the convolution of the functions $\tilde{H}_{\mathbf{g}}(\mathbf{k})$ with the reciprocal lattice vectors $\mathbf{g}$. For the case of a perfect crystal the Fourier transform of the image will be non-zero only at the Bragg positions and the functions $\tilde{H}_{\mathbf{g}}(\mathbf{k})$ will be delta functions of height $H_{\mathbf{g}}$. If there are variations in the image, however, the Fourier transform will be non-zero between the Bragg positions. For values of $\mathbf{k}$ within one Brillouin zone of the position $\mathbf{g}$ we can write:

$$\tilde{I}(\mathbf{k}) = \tilde{H}_{\mathbf{g}}(\mathbf{k} - \mathbf{g}). \quad (6)$$

This can be made explicit by introducing a masking function $\tilde{M}(\mathbf{k})$ such that

$$\tilde{M}(\mathbf{k}) = 1 \quad \text{inside the first Brillouin zone}, \quad (7)$$

$$\tilde{M}(\mathbf{k}) = 0 \quad \text{outside the first Brillouin zone}. \quad (8)$$

We can therefore define $H_{\mathbf{g}}(\mathbf{r})$ uniquely by the following equation:

$$\tilde{H}_{\mathbf{g}}(\mathbf{k}) = \tilde{I}(\mathbf{k} + \mathbf{g}) \tilde{M}(\mathbf{k}). \quad (9)$$

In this way, the Fourier transform of the image intensity, given by Eq. (5), is mapped out in terms of the functions $\tilde{H}_{\mathbf{g}}(\mathbf{k})$.

2.1. Interpretation of geometric phase images

The Fourier components, $H_{\mathbf{g}}$, of the image of a perfect crystal, $I(\mathbf{r})$, describe the amplitude and phase of the corresponding sets of lattice fringes $\mathbf{g}$. Writing Eq. (1) in terms of the amplitude, $A_{\mathbf{g}}$, and phase, $P_{\mathbf{g}}$, as defined by Eq. (2), and using that for a real image there is conjugate symmetry between the Fourier components, the following equation is obtained:

$$I(\mathbf{r}) = A_0 + \sum_{\mathbf{g} > 0} 2A_{\mathbf{g}} \cos\{2\pi \mathbf{g} \cdot \mathbf{r} + P_{\mathbf{g}}\} \quad (10)$$

which is simply the alternative way of writing a Fourier transform of a real function. The image of a particular set of lattice fringes $B_{\mathbf{g}}(\mathbf{r})$ is therefore given by

$$B_{\mathbf{g}}(\mathbf{r}) = 2A_{\mathbf{g}} \cos\{2\pi \mathbf{g} \cdot \mathbf{r} + P_{\mathbf{g}}\} \quad (11)$$

which is the image which would be produced by Bragg filtering of the original image (placing a mask around the positions $\pm \mathbf{g}$ in the Fourier transform). In the presence of variations in the image fringes, it is easy to show that the conjugate symmetry remains:

$$H_{-\mathbf{g}}(\mathbf{r}) = H_{\mathbf{g}}^*(\mathbf{r}). \quad (12)$$

The Bragg filtered images produced using the Brillouin zone mask $\tilde{M}(\mathbf{k})$, defined by Eqs. (7) and (8), will therefore be given by

$$B_{\mathbf{g}}(\mathbf{r}) = 2A_{\mathbf{g}}(\mathbf{r}) \cos\{2\pi \mathbf{g} \cdot \mathbf{r} + P_{\mathbf{g}}(\mathbf{r})\}. \quad (13)$$

This was the starting point used previously to introduce the concept of geometric phase images [16]. Here we shall concentrate on the phase images $P_{\mathbf{g}}(\mathbf{r})$, the interpretation of the amplitude images $A_{\mathbf{g}}(\mathbf{r})$ has been treated elsewhere [17].

2.1.1. Change in reciprocal lattice vector

Let us assume that the reciprocal lattice vector in the image differs from that of the reference lattice. In that case:

$$\mathbf{g} \rightarrow \mathbf{g} + \Delta\mathbf{g}, \quad (14)$$

where $\Delta\mathbf{g}$ is the difference in reciprocal lattice vector from the reference lattice $\mathbf{g}$. The perfect set of fringes described by Eq. (11) will become

$$B_g(\mathbf{r}) = 2A_g \cos\{2\pi\mathbf{g} \cdot \mathbf{r} + 2\pi\Delta\mathbf{g} \cdot \mathbf{r} + P_g\}. \quad (15)$$

Comparing this equation with Eq. (13) we can see that the phase as a function of position will be given by

$$P_g(\mathbf{r}) = 2\pi \Delta\mathbf{g} \cdot \mathbf{r}, \quad (16)$$

where the arbitrary constant phase term P_g has been ignored. A difference in reciprocal lattice vector therefore produces a uniform ramp in the phase image. Taking the gradient of Eq. (16) gives the following result:

$$\nabla P_g(\mathbf{r}) = 2\pi \Delta\mathbf{g}. \quad (17)$$

We make the assumption that in the presence of a varying reciprocal lattice vector, this equation holds true and the gradient in the phase gives the local deviation, $\Delta\mathbf{g}(\mathbf{r})$, from the reference lattice, $\mathbf{g}$. In the limit of a slowly varying distortion of the lattice this must be true, as has been shown. The effective lateral resolution of the method is discussed later.

2.1.2. Displacement fields

Following an argument first developed for the theory of dynamical scattering from defects [20], in the presence of a displacement field, $\mathbf{u}$:

$$\mathbf{r} \rightarrow \mathbf{r} - \mathbf{u} \quad (18)$$

and the set of perfect fringes, Eq. (11) becomes

$$B_g(\mathbf{r}) = 2A_g \cos\{2\pi\mathbf{g} \cdot \mathbf{r} - 2\pi\mathbf{g} \cdot \mathbf{u} + P_g\}. \quad (19)$$

The maximum of the fringes will therefore be displaced by a vector $\mathbf{u}$ with respect to their initial position. Comparing this equation with Eq. (13) we can see that

$$P_g(\mathbf{r}) = -2\pi\mathbf{g} \cdot \mathbf{u} \quad (20)$$

ignoring the arbitrary constant phase P_g . This equation was derived in the limit of a uniform displacement of all the fringes and is *exact*; we again extend this result to include varying displacement fields, $\mathbf{u}(\mathbf{r})$. This is very similar to the deformation field approach [21] which describes the effect of a displacement field by including phase terms in the Fourier transform of the perfect crystal potential. As for the previous case, the relation will break down for rapidly varying distortions. However, the results from different lattice fringes should be consistent with each other. We can define a consistency rule in the following way:

$$P_g(\mathbf{r}) = P_{g-g'}(\mathbf{r}) + P_{g'}(\mathbf{r}) \quad (21)$$

which can easily be verified by substituting into Eq. (20). The interpretation of the phase images in terms of a displacement field can therefore be tested by checking the results from several different sets of lattice fringes.

The two interpretations of the phase images, described by Eqs. (16) and (20), are equivalent. The gradient of the displacement field gives the local fringe spacing which, for small gradients, gives the change in the reciprocal lattice vector, as will be shown later.

2.2. Construction of the phase image

The power spectrum of the image intensity $I(\mathbf{r})$ is first calculated after applying a suitable mask (von Hann or other) in real space to avoid streaking. One of the peaks in the power spectrum is then chosen as the reciprocal lattice vector $\mathbf{g}$, the components corresponding to the position of the intensity maximum (this can be calculated to sub-pixel accuracy as the centre of mass of the intensities in the peak). The Fourier transform, $\tilde{I}(\mathbf{k})$, of the original image is then calculated and the mask $\tilde{M}(\mathbf{k})$ applied to the position $\mathbf{g}$. (Note that unlike Bragg filtering, the full Fourier transform must be calculated and not just the half-plane, i.e. the mask is not applied to the $-\mathbf{g}$ position.) This is equivalent to selecting one of the terms in Eq. (3) and on taking the inverse Fourier transform the resulting complex image $H_g(\mathbf{r})$ will be

$$H_g(\mathbf{r}) = H_g(\mathbf{r}) \exp\{2\pi i \mathbf{g} \cdot \mathbf{r}\}. \quad (22)$$

Writing $H_g(\mathbf{r})$ in terms of the amplitude, $A_g(\mathbf{r})$, and phase, $P_g(\mathbf{r})$, as defined by Eq. (2) gives

$$H'_g(\mathbf{r}) = A_g(\mathbf{r}) \exp\{2\pi i \mathbf{g} \cdot \mathbf{r} + i P_g(\mathbf{r})\}. \quad (23)$$

The Bragg-filtered image intensity, $B_g(\mathbf{r})$, amplitude, $A_g(\mathbf{r})$, and phase, $P_g(\mathbf{r})$ images are then calculated from this image in the following way:

$$B_g(\mathbf{r}) = 2\Re[H'_g(\mathbf{r})], \quad (24)$$

$$A_g(\mathbf{r}) = \text{Mod}[H'_g(\mathbf{r})], \quad (25)$$

$$P_g(\mathbf{r}) = \text{Phase}[H'_g(\mathbf{r})] - 2\pi \mathbf{g} \cdot \mathbf{r}, \quad (26)$$

$$P'_g(\mathbf{r}) = \text{Phase}[H'_g(\mathbf{r})], \quad (27)$$

where $\Re$ denotes the real part and $P'_g(\mathbf{r})$ is what can be defined as the *raw-phase* image. The raw-phase image, $P'_g(\mathbf{r})$, is useful because the local reciprocal lattice vector, $\mathbf{g}(\mathbf{r}) = \mathbf{g} + \Delta\mathbf{g}(\mathbf{r})$, can be determined directly by differentiation (cf. Eq. (17)):

$$\nabla P'_g(\mathbf{r}) = 2\pi \mathbf{g}(\mathbf{r}). \quad (28)$$

The phase image, $P_g(\mathbf{r})$, is obtained from the raw-phase image by subtracting the factor $2\pi \mathbf{g} \cdot \mathbf{r}$, as described in Eq. (26), followed by a renormalisation between $\pm \pi$.

In practice, the mask used in reciprocal space, $\tilde{M}(\mathbf{k})$, is usually not the size of one Brillouin zone, as defined by Eqs. (7) and (8), but rather a circular or Gaussian mask. In order to reduce noise and to smooth the resulting images, a Gaussian mask has been used in the examples shown in this paper:

$$\tilde{M}(\mathbf{k}) = \exp\left\{-4\pi \frac{k^2}{g^2}\right\}. \quad (29)$$

Appropriate ways to choose the parameters of such masks and the relation with Gabor filtering [22] have been discussed elsewhere [17].

The method of calculating the phase images from the Fourier transform is identical to that used in optical interferometry [19]. The method is also very similar to that used in holography for reconstructing the amplitude and phase of the electron wavefunction with the position of the reference wave chosen as the Bragg position $\mathbf{g}$. The phase which is produced corresponds to the positions of lattice fringes in the image and has been called *geometric* [16] to avoid confusion with the phase of the electron wavefunction [23].

All the image processing was carried out using routines written with SEMPER [24] but equivalent routines have also been implemented for DigitalMicrograph [25].

3. Determination of the displacement field

The phase image, $P_g(\mathbf{r})$, gives the component of the displacement field, $\mathbf{u}(\mathbf{r})$, in the direction of the reciprocal lattice vector, $\mathbf{g}$. By combining the information from two sets of lattice fringes the vectorial displacement field can be calculated (provided that the reciprocal lattice vectors, $\mathbf{g}_1$ and $\mathbf{g}_2$ are non-colinear). From Eq. (20) we have

$$P_{g1}(\mathbf{r}) = -2\pi \mathbf{g}_1 \cdot \mathbf{u}(\mathbf{r}) \quad (30)$$

$$= -2\pi\{g_{1x}u_x(\mathbf{r}) + g_{1y}u_y(\mathbf{r})\}, \quad (31)$$

$$P_{g2}(\mathbf{r}) = -2\pi \mathbf{g}_2 \cdot \mathbf{u}(\mathbf{r}) \quad (32)$$

$$= -2\pi\{g_{2x}u_x(\mathbf{r}) + g_{2y}u_y(\mathbf{r})\}, \quad (33)$$

where g_{1x} and g_{1y} are the k_x and k_y components ($\mathbf{k}$ being the variable in reciprocal space) of the vector $\mathbf{g}_1$, and $u_x(\mathbf{r})$ and $u_y(\mathbf{r})$ are the x and y components of the displacement field at the position $\mathbf{r} = (x, y)$ in the image. A note concerning digital images can be found in Appendix A. The displacement field with respect to the two-dimensional lattice defined by the vectors $\mathbf{g}_1$ and $\mathbf{g}_2$ is given by solving these simultaneous equations. It is instructive to carry this out by writing Eqs. (31) and (33) in matrix form:

$$\begin{pmatrix} P_{g1} \\ P_{g2} \end{pmatrix} = -2\pi \begin{pmatrix} g_{1x} & g_{1y} \\ g_{2x} & g_{2y} \end{pmatrix} \begin{pmatrix} u_x \\ u_y \end{pmatrix}, \quad (34)$$

where the relationship between the phase and displacement field as a function of position in the image has now been made implicit. The displacement field can be calculated by taking the inverse of the matrix containing the reciprocal lattice vectors:

$$\begin{pmatrix} u_x \\ u_y \end{pmatrix} = -\frac{1}{2\pi} \begin{pmatrix} g_{1x} & g_{1y} \\ g_{2x} & g_{2y} \end{pmatrix}^{-1} \begin{pmatrix} P_{g1} \\ P_{g2} \end{pmatrix}. \quad (35)$$

The inverse of this matrix is easy to calculate but to show its significance, we now introduce the

vectors $\mathbf{a}_1$ and $\mathbf{a}_2$ which correspond to the lattice in real space defined by the reciprocal lattice vectors $\mathbf{g}_1$ and $\mathbf{g}_2$. Let

$$A = \begin{pmatrix} a_{1x} & a_{2x} \\ a_{1y} & a_{2y} \end{pmatrix} \quad \text{and} \quad G = \begin{pmatrix} g_{1x} & g_{2x} \\ g_{1y} & g_{2y} \end{pmatrix}. \quad (36)$$

It is easy to show (see Appendix B) that $G^T = A^{-1}$ and Eq. (35) becomes

$$\begin{pmatrix} u_x \\ u_y \end{pmatrix} = -\frac{1}{2\pi} \begin{pmatrix} a_{1x} & a_{2x} \\ a_{1y} & a_{2y} \end{pmatrix} \begin{pmatrix} P_{g1} \\ P_{g2} \end{pmatrix}. \quad (37)$$

Returning to a vectorial representation the following result is obtained:

$$\mathbf{u}(\mathbf{r}) = -\frac{1}{2\pi} [P_{g1}(\mathbf{r})\mathbf{a}_1 + P_{g2}(\mathbf{r})\mathbf{a}_2]. \quad (38)$$

It is this simple relation that will be used to calculate the vectorial displacement field from two-phase images. The equivalent relation using Miller indices and in terms of the crystal lattice is derived in Appendix C.

4. Determination of the strain field

The local distortion of the lattice is given by the gradient of the displacement field and can be described by the 2×2 matrix, e , defined as follows [26]:

$$e = \begin{pmatrix} e_{xx} & e_{xy} \\ e_{yx} & e_{yy} \end{pmatrix} = \begin{pmatrix} \frac{\partial u_x}{\partial x} & \frac{\partial u_x}{\partial y} \\ \frac{\partial u_y}{\partial x} & \frac{\partial u_y}{\partial y} \end{pmatrix}. \quad (39)$$

This matrix can be separated into two terms, the symmetric term, ε , and the antisymmetric term, ω , defined as follows:

$$\varepsilon = \frac{1}{2} \{e + e^T\}, \quad (40)$$

$$\omega = \frac{1}{2} \{e - e^T\}, \quad (41)$$

where T denotes the transpose of the matrix. The strain is given by ε and the local rigid rotation by ω , for small distortions. For the more general case, the distortions can still be described in terms of a rotation matrix and a symmetric matrix (see, e.g., Ref. [8]).

The derivative is calculated numerically from the images. Since the images are already smoothed by the limited size of the mask used in reciprocal space to create the phase images, we have found it sufficient to carry out the calculation of the derivative by subtracting the values in neighbouring pixels, pair-by-pair across the image. To avoid an accumulation of errors, it is best to carry out this procedure on the phase images, rather than the displacement images, and by substituting Eq. (37) into Eq. (39) we find that

$$e = -\frac{1}{2\pi} \begin{pmatrix} a_{1x} & a_{2x} \\ a_{1y} & a_{2y} \end{pmatrix} \begin{pmatrix} \frac{\partial P_{g1}}{\partial x} & \frac{\partial P_{g1}}{\partial y} \\ \frac{\partial P_{g2}}{\partial x} & \frac{\partial P_{g2}}{\partial y} \end{pmatrix}. \quad (42)$$

As will be seen later, the phase images have apparent discontinuities where the phase changes abruptly from $-\pi$ to $+\pi$. These produce discontinuities in the derivatives which have no physical significance and result simply from the fact that the phase is normalised to the range $\pm\pi$. The problem is solved by calculating the derivative from the complex image $\exp\{iP_g(\mathbf{r})\}$ (see Appendix D). Another possibility is to “unwrap” the phase, i.e. to eliminate the phase discontinuities by adding multiples of 2π where necessary, before taking the derivative. However, there is no unique solution for unwrapping the phase in two-dimensions and sophisticated methods of analysis are required (for a recent review see, e.g., Ref. [27]).

5. Determination of the local crystal lattice

The logical conclusion of the method is to determine the crystal lattice locally across the image. The reference reciprocal lattice is described by the matrix G , as defined by Eq. (36). Let $G(\mathbf{r})$ be the local reciprocal lattice which deviates from the reference lattice G by $\Delta G(\mathbf{r})$:

$$G(\mathbf{r}) = G + \Delta G(\mathbf{r}), \quad (43)$$

where

$$\Delta G = \begin{pmatrix} \Delta g_{1x} & \Delta g_{2x} \\ \Delta g_{1y} & \Delta g_{2y} \end{pmatrix}. \quad (44)$$

The deviations from the local reciprocal lattice are given by the phase images. Using Eq. (17) we have

$$\Delta G = \frac{1}{2\pi} \begin{pmatrix} \frac{\partial P_{g1}}{\partial x} & \frac{\partial P_{g2}}{\partial x} \\ \frac{\partial P_{g1}}{\partial y} & \frac{\partial P_{g2}}{\partial y} \end{pmatrix}. \quad (45)$$

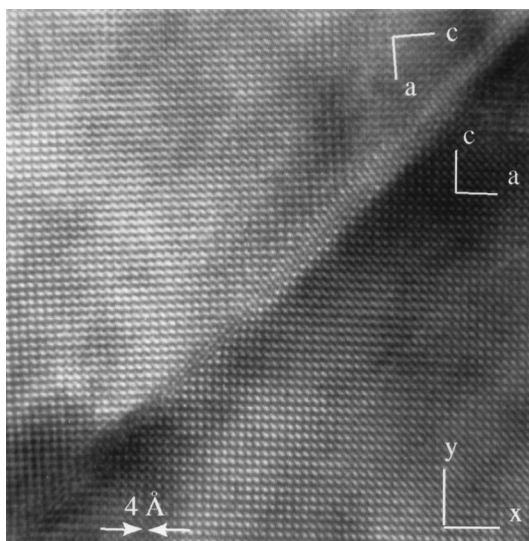
The local reciprocal lattice $G(\mathbf{r})$ can therefore be calculated point by point in the image. Practically, $G(\mathbf{r})$ can be calculated directly from the raw-phase images, $P'_g(\mathbf{r})$, using Eq. (28). Using the relation $G^T = A^{-1}$ derived in Appendix B, the matrix $A(\mathbf{r})$ can be calculated thus giving the local lattice in real space as a function of position in the image. Appendix E uses this result to derive the deformation matrix relating the reference lattice and the distorted lattice.

6. Application of the technique: domain walls in PbTiO_3

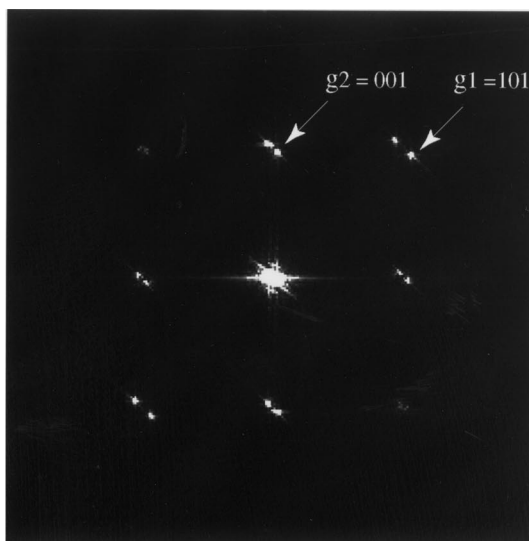
To illustrate the technique we shall study the displacement and strain fields associated with do-

main walls in PbTiO_3 . The material undergoes a phase transition at about 480°C from a high temperature paraelectric cubic phase to a non-centrosymmetric tetragonal phase which is ferroelastic and ferroelectric. The domains are defined by the relative direction of the a - and c -axis. The elastic and electric properties of the material are strongly influenced by the structure of the interfaces between domains so it is important to determine whether the tetragonality (c/a ratio) varies across the boundary and how abruptly the rotation of the lattice takes place. Such materials have been previously studied by HREM using the peak-finding method [29–31] and a preliminary study using phase images has been presented which includes a brief comparison with the peak-finding and optical moiré techniques [32].

Fig. 1a shows an $[0\ 1\ 0]$ high-resolution electron microscope image, taken on a Philips CM30 (accelerating voltage 300 kV, C_s 1.2 mm), of a single 90° domain wall across which the a - and c -axis interchange (an a - a domain wall). The specimen was prepared according to the method described in Ref. [28]. The negative was digitised using a flatbed Nikon LS 4500 AF scanner and the resulting image



(a)



(b)

Fig. 1. Ferroelastic-ferroelectric a - a domain wall in PbTiO_3 : (a) $[0\ 1\ 0]$ high-resolution image; (b) power spectrum of the image, reference lattice $\mathbf{g}_1 = 101$ and $\mathbf{g}_2 = 001$ marked.

was 512 square with a sampling density of 0.047 nm/pixel. Peak-finding methods typically require a much higher sampling density which has obvious practical disadvantages compared with that needed for the phase method.

6.1. Calculation of the phase images

The two most intense spots in the power spectrum (Fig. 1b), $g_1 = 101$ and $g_2 = 001$, were chosen to calculate the displacement field so as to produce results with the best signal-to-noise ratio. These spots correspond to the region on the right-hand side of the domain wall, which will serve as the reference lattice.

A Gaussian mask was placed around the $g_1 = 101$ position in the Fourier transform of the image as described in Section 2.2. An inverse-Fourier transform was then performed to create the complex image $H'_{101}(r)$, described in Eq. (22). This image was used to calculate the Bragg filtered image, $B_{101}(r)$, the phase image, $P_{101}(r)$, and the raw-phase image, $P'_{101}(r)$, according to the Eqs. (24), (26) and (27) respectively. The amplitude images are not analysed here, though for other examples, they can provide valuable information [17,33]. A similar procedure was carried out for the other reciprocal lattice vector, $g_2 = 001$, and the results are shown in Fig. 2.

The Bragg filtered images shown in Fig. 2a and Fig. 2b are very similar to the raw phase images, $P'_g(r)$, shown in Fig. 2c and Fig. 2d. The maxima of the fringes occur when the raw-phase images are π . The sudden changes in grey level are simply due to the fact that for all phase images the values are limited between $\pm \pi$. If the phase decreases just below $-\pi$ (black) it is renormalised to just below $+\pi$ (white). In this way, we see the fringes. The phase images, $P_g(r)$, shown in Fig. 2e and Fig. 2f are obtained from $P'_g(r)$ by subtracting the reference lattice phase $2\pi g \cdot r$, as described in Eq. (26), and renormalising the result between $\pm \pi$.

One of the advantages of the phase images over the Bragg filtered images is that variations from the reference lattice are much easier to see. On the right-hand side of the domain wall the phase is approximately zero as this corresponds to the area of crystal chosen as the reference lattice. To the left

of the domain wall, the phase shows a significant gradient. As has been explained in Section 2.1, this means that the local reciprocal lattice vector is different from the reference lattice, by an amount Δg given by Eq. (17). The direction of the gradient corresponds to the direction of Δg which corresponds well with the position of the neighbouring spot in the diffractogram (Fig. 1b). The gradient of the 101 phase is greater than that of the 001 phase as the distance separating the neighbouring spots is also greater (approximately twice as much). For the 101 fringes, Δg_{101} is perpendicular to the reference lattice vector g_{101} which means that the lattice fringes are rotated with respect to the right-hand side of the domain wall, which can be verified by looking at the lattice fringes themselves in Fig. 2a (or Fig. 2c). For the 001 fringes, the lattice spacing also varies, although in the image, Fig. 2b (or Fig. 2d) it is the rotation which is most visible.

A practical aspect of the phase images is therefore that they help the interpretation and indexation of the periodicities present in the power spectrum of the image. In the power spectrum of the image (Fig. 1b) each spot was split into two. In order to determine which spot corresponded to which side of the domain wall it was sufficient to calculate the phase image having chosen one of the two. The region showing the constant phase identifies the reference lattice. Indeed, this was how the spots g_1 and g_2 marked in Fig. 1b were identified as corresponding to the right-hand side of the domain wall.

Another aspect of the images is that the 001 phase changes abruptly across the domain wall. This can be seen more clearly in Fig. 3b where a line profile of the phase has been plotted as a function of position across the domain wall from right to left. (To reduce noise, the profile has been averaged along the interface.) The step in the phase gives the rigid-body displacement of the fringes via Eq. (20). In this case, the phase change is approximately π which corresponds to a translation of half a lattice fringe spacing. Indeed, the 001 fringes in Fig. 2b can be seen to be discontinuous at the domain boundary. The step in the 101 phase (Fig. 3a) is much smaller as these fringes are almost continuous (see Fig. 2a).

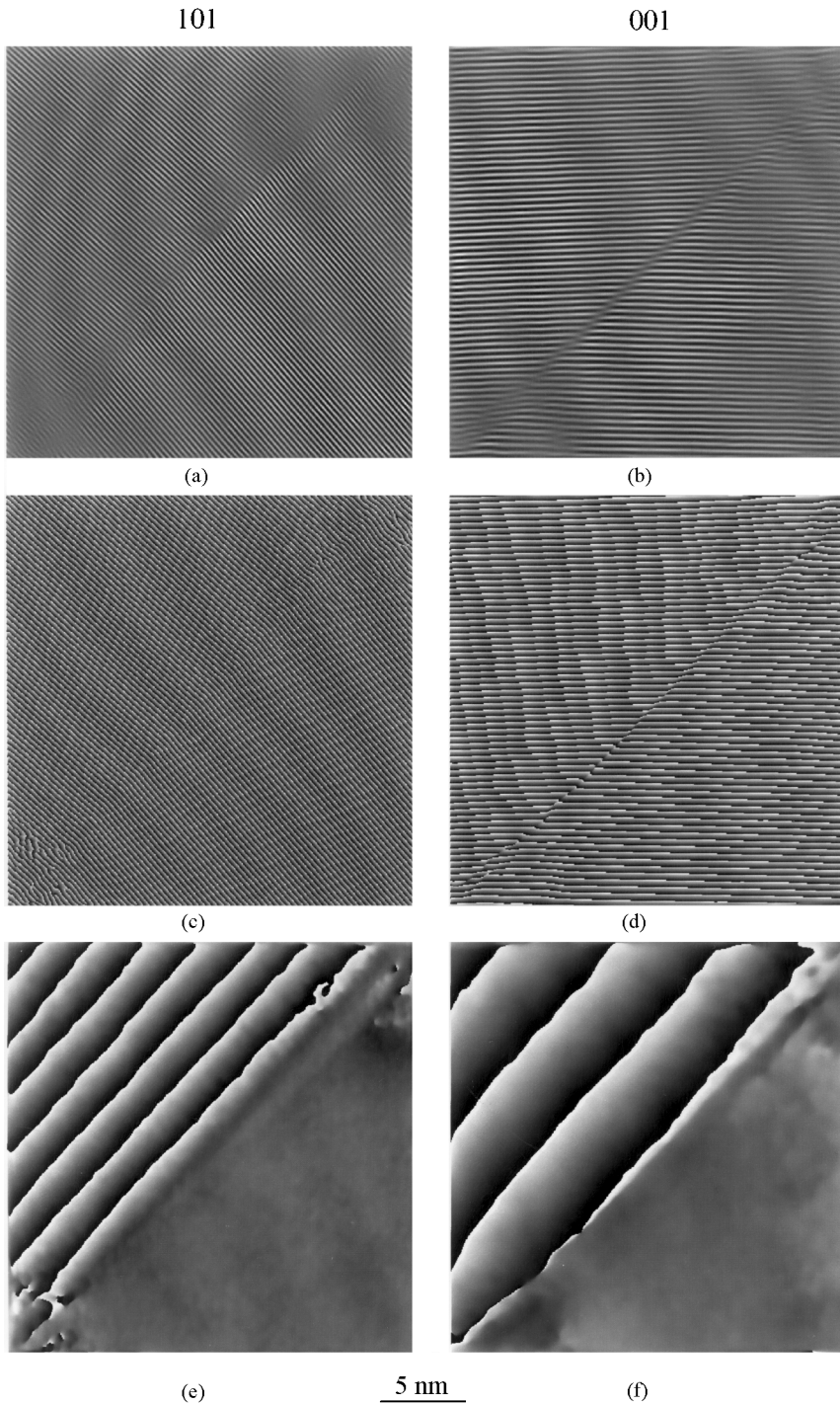


Fig. 2. Method of construction of the phase images: (a) 101 Bragg filtered image, $B_{101}(r)$; (b) 001 Bragg filtered image, $B_{001}(r)$; (c) raw phase image, $P'_{101}(r)$; (d) raw phase image, $P'_{001}(r)$; (e) 101 phase image, $P_{101}(r)$; (f) 001 phase image, $P_{001}(r)$. For the phase images black = $-\pi$ white = π .

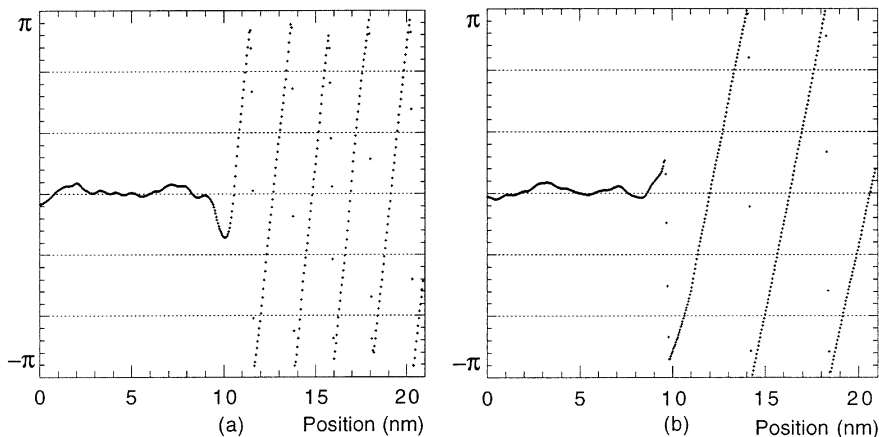


Fig. 3. Phase profiles across the domain wall with the reference lattice on the left: (a) 101 phase profile; (b) 001 phase profile.

6.2. Determination of the displacement field

The two reciprocal lattice vectors $\mathbf{g}_1 = 101$ and $\mathbf{g}_2 = 001$ define a lattice in real space given by $\mathbf{a}_1 = [100]$ and $\mathbf{a}_2 = [\bar{1}01]$. It can be easily verified that these vectors satisfy the relations in Appendix B. The displacement field is then given directly by Eq. (38):

$$\mathbf{u}(\mathbf{r}) = -\frac{1}{2\pi} \{P_{101}(\mathbf{r})[100] + P_{001}(\mathbf{r})[\bar{1}01]\}. \quad (46)$$

This equation can be used to calculate the rigid-body translation $\mathbf{u}_r$ across the boundary. The height of the discontinuity in the phase profiles shown previously in Fig. 3 was measured giving the following result:

$$\mathbf{u}_r(\mathbf{r}) = (0.12 \pm 0.03)[100] + (0.53 \pm 0.03)[\bar{1}01]. \quad (47)$$

The errors have been estimated from the phase profiles and are due to the fact that even in the reference lattice the displacement field is not constant. To show the vector field, we can describe $\mathbf{u}(\mathbf{r})$ in terms of the x and y -components, $u_x(\mathbf{r})$ and $u_y(\mathbf{r})$. Rather than choosing the axes to coincide with the axes of the image, which depend on the digitisation, we have chosen the x -axis to coincide with the (100) direction and the y -axis with the (001) direction of the reference lattice, as shown on Fig. 1a,

and to have unit lengths corresponding to the a and c lattice parameters, respectively. The displacement field will then be expressed in terms of fractions of unit cell vectors. Eq. (37) then takes on a very simple form:

$$\begin{pmatrix} u_x \\ u_y \end{pmatrix} = -\frac{1}{2\pi} \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} P_{101} \\ P_{001} \end{pmatrix}. \quad (48)$$

This equation is carried out pixel by pixel across the images and so is a form of image algebra (for a brief introduction, see e.g., Ref. [34]). Fig. 4 shows a representation of Eq. (48) with the functions $P_g(\mathbf{r})$ and $\mathbf{u}(\mathbf{r})$ replaced by the corresponding images.

The x - and y -components of the displacement field $\mathbf{u}(\mathbf{r})$ so obtained are shown in Fig. 5a and b, respectively. To the right of the domain wall, the displacement field is zero on average because this was the region of crystal used to define the reference lattice. The size of the displacements in this region are useful in assessing the error in the measured values for the displacement field. In this case, the variations are small (standard deviation of 0.02 nm in both x and y), presumably due to noise, thickness variations and slight changes in orientation of the crystal, except towards the top right and bottom left of the figure. Measurements of the tetragonality were therefore taken from the middle section of the domain wall where the variations in the displacement field in the region of the reference lattice are

$$\begin{aligned}
 \text{(a)} \quad \begin{pmatrix} \mathbf{u}_x \\ \mathbf{u}_y \end{pmatrix} &= -\frac{1}{2\pi} \begin{pmatrix} \mathbf{a}_{1x} & \mathbf{a}_{2x} \\ \mathbf{a}_{1y} & \mathbf{a}_{2y} \end{pmatrix} \begin{pmatrix} \mathbf{P}_{g1} \\ \mathbf{P}_{g2} \end{pmatrix} \\
 \text{(b)} \quad \begin{pmatrix} \text{Image} \\ \text{Image} \end{pmatrix} &= -\frac{1}{2\pi} \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \text{Image} \\ \text{Image} \end{pmatrix}
 \end{aligned}$$

Fig. 4. Calculation of the displacement field: (a) matrix equation; (b) image equation, with images replacing the functions of position shown in (a). The x -axis is parallel to the 100 direction and the y -axis is parallel to the 001 direction of the reference lattice. The displacement field is given as fractions of unit cell vectors where black = -0.5 white = 0.5 .

the smallest. The images of the displacement fields are therefore very useful, in a practical sense, for determining regions of the crystal where reliable measures can be carried out.

The apparently abrupt changes in the displacement field are simply where the corresponding phase images changed from $-\pi$ to $+\pi$. The displacement field is always measured with respect to a reference lattice, when the displacement vector increases beyond half a unit cell it is automatically renormalised to minus one-half a unit cell.

6.3. Calculation of the local lattice distortion

To analyse the local tetragonality and rotation of the lattice the local distortion of the lattice, $e(\mathbf{r})$, was calculated from the displacement field $\mathbf{u}(\mathbf{r})$ according to Eq. (39). Practically, this was carried out directly from the phase images using Eq. (42). This matrix was then separated into the symmetric strain matrix, $\varepsilon(\mathbf{r})$ and rotation matrix, $\omega(\mathbf{r})$. The results are shown in Figs. 6 and 7. Each matrix element is an image, the values depending on the position $\mathbf{r}$.

The definition of the strain is with respect to the undistorted lattice. Since we do not know the unstrained lattice, we are measuring the distortion of the lattice with respect to the chosen reference lattice. The strain matrix, shown in Fig. 6, therefore

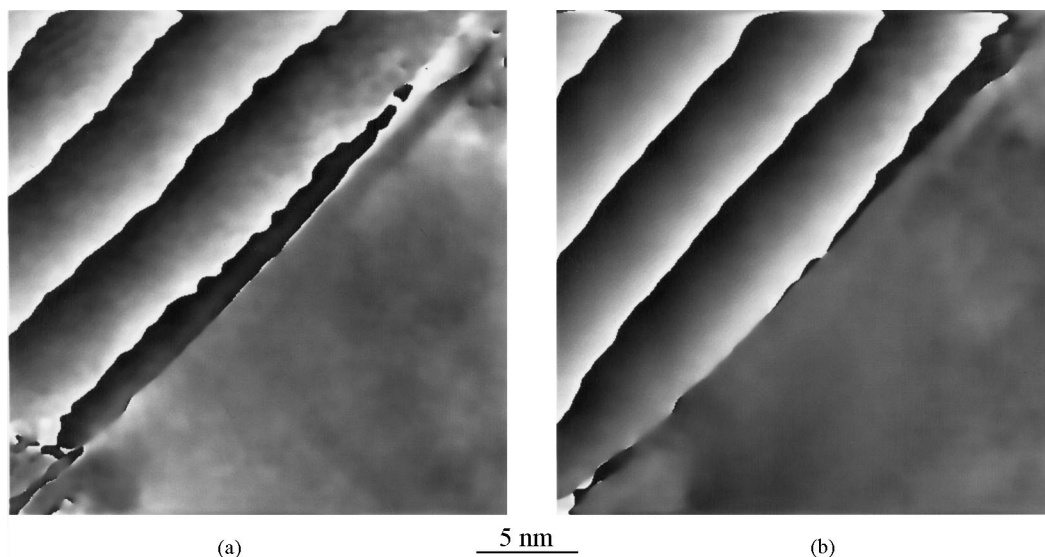


Fig. 5. Displacement field: (a) x -component of the displacement field, $u_x(\mathbf{r})$; (b) y -component of the displacement field $u_y(\mathbf{r})$. The x -axis is parallel to the 100 direction and the y -axis is parallel to the 001 direction of the reference lattice. The displacement field is given as fractions of unit cell vectors where black = -0.5 white = 0.5 .

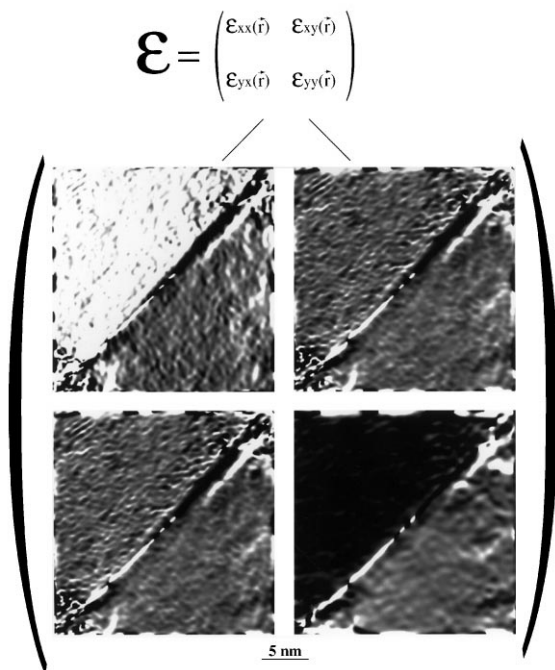


Fig. 6. Strain matrix $\varepsilon(r)$. Black = -0.06 white = 0.06 .

describes how the crystal lattice varies across the image – and should only be understood in these terms.

To the right of the domain wall, $\varepsilon_{xx}(\mathbf{r})$, is zero on average (the grey contrast) – this is the reference lattice – to the left it is positive and equal to $6.1 \pm 0.4\%$ on average. This means that the lattice spacing has increased with respect to the reference lattice which corresponds to the a - and c -axis interchanging. For the lattice planes along the y -axis the opposite occurs with the c -spacing in the reference lattice becoming the a -spacing to the left of the domain wall ($\langle \varepsilon_{yy} \rangle = -5.8 \pm 0.4\%$). The off-diagonal elements ($\varepsilon_{xy}(\mathbf{r})$ and $\varepsilon_{yx}(\mathbf{r})$), which for small distortions correspond to a shear, are on average zero (grey contrast).

There seems to be evidence for oscillations in the strain field close to the domain wall as has been recently reported for grain boundaries in Mo [35]. It is possible, however, that the objective lens distorts the image because of the abrupt change in the structure at the domain wall. These distortions can also vary along the boundary because of changes in

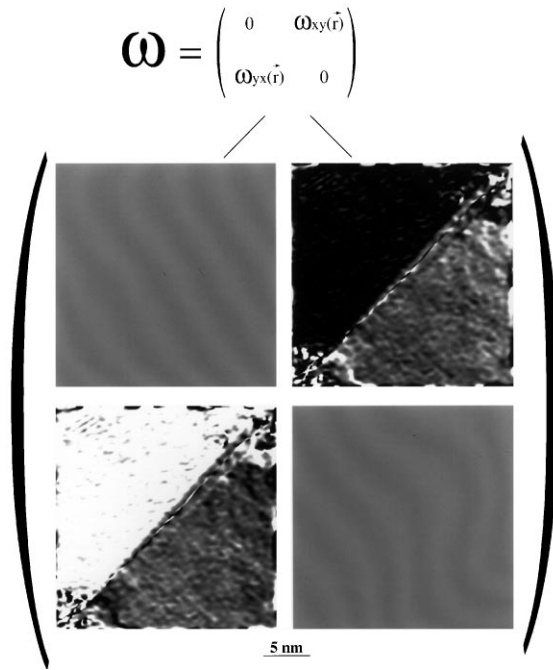


Fig. 7. Rotation matrix $\omega(r)$. Black = -0.06 white = 0.06 .

the specimen thickness. A comparison with image simulations using different structural models, as was carried out for the Mo grain boundary, would be necessary to be sure.

The off-diagonal elements in the matrix $\omega(\mathbf{r})$, shown in Fig. 7, can be interpreted directly in terms of a rotation of the lattice. For small values, the rotation angle in radians is equal to ω_{yx} , the average value being $3.5 \pm 0.1^\circ$ of rotation from one side of the boundary to the other. The rotation of the lattice appears to occur abruptly at the domain wall and without the fluctuations seen in the strain images. This could be that the information concerning a rotation is perpendicular to the lattice vector in reciprocal space whereas for a change in lattice parameter, the information is distributed in a radial direction. The transfer function of the objective lens varies much less around the optic axis (and in the absence of misalignments not at all) than radially. Rotations are therefore much more likely to be imaged faithfully by the lens. This could explain the difference between the images of the strain and the rotation.

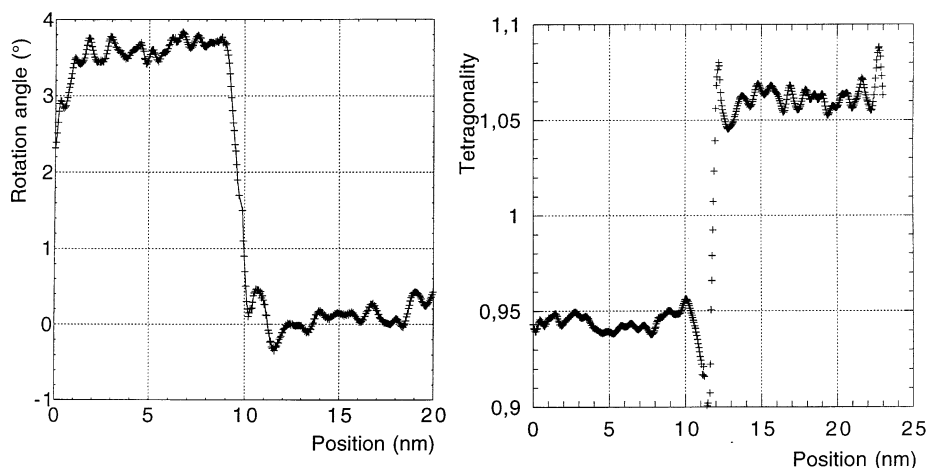


Fig. 8. Line profiles across the domain boundary with the reference lattice on the right: (a) local rotation of the lattice; (b) variation in the tetragonality $T = c/a$, a and c correspond to the lattice parameters measured along the x and y -axes of the reference lattice respectively.

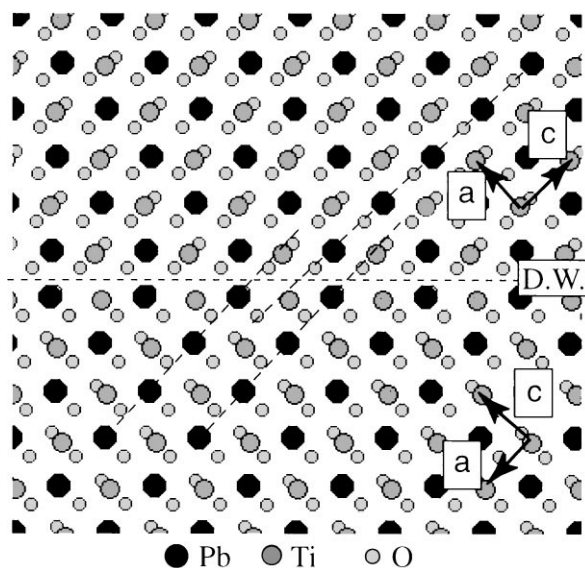


Fig. 9. Model for the domain wall structure consistent with the measured parameters.

To show the change in the lattice more clearly, line profiles were taken across the interface from left to right for the rotation and the strain matrix. The results, which have been averaged along the interface to reduce noise, are shown in Fig. 8. The tetragonality T has been defined as the ratio of the

lattice parameters measured along the a and c axis of the reference lattice. Theoretically, the angle made by the domain wall is given by $90 - 2 \arctan(a/c)^\circ = 3.4^\circ$ which corresponds well with the measured value of $3.5 \pm 0.1^\circ$. The interface is relatively abrupt, taking place over less than 1 nm, which agrees with results obtained elsewhere for an a - a domain wall [29]. A model for the domain wall structure which is consistent with these measurements, including the rigid body displacement, is presented in Fig. 9.

7. Conclusions

Fourier filtering is an image processing technique often used in HREM either to improve image contrast by placing masks around the Bragg spots in the Fourier transform or, less commonly, by selecting individual lattice fringes to help the interpretation of image contrast. Here, we have shown how the analysis of lattice fringes can be placed in a coherent theoretical framework and how the analysis can be made quantitative by introducing the concept of “local” values of Fourier components.

Images can be obtained of the amplitude and phase of the lattice fringes as a function of position

in the image. The phase images can be directly related to the displacement field distorting the fringes with respect to the reference lattice. By choosing two independent sets of lattice fringes, the two-dimensional displacement field can be calculated. The relationship between phase and displacement field has a particularly simple form when written in terms of the basis vectors of the lattice in real space. Once the displacement field has been calculated, the calculation of the local deformation of the lattice is trivial.

The phase images themselves are extremely quick and easy to calculate as they involve essentially only two Fourier transforms. The computing time is therefore a matter of a few seconds depending on the size of the digitised region analysed. Practically, the only requirement on the image is that there be identifiable Bragg spots in the Fourier transform of the image. It is also often the case that due to misalignments only a few fringes are of good contrast in the image. This means that the phase method can select the spots with the best contrast to calculate the displacement field. Another advantage which should not be overlooked is that a continuous displacement field is produced. This greatly facilitates the analysis and mathematical manipulation of the images and also provides a link with elastic continuum model theories of deformations.

More work needs to be carried out in order to determine how closely lattice fringe positions correspond to atomic plane positions. The empirical response is that other lattice fringes in the image could be used to calculate the displacement fields. The results could then be compared with previous calculations and any disparities used as a measure of consistency. Where the factors detailed in the introduction are too important to ignore, simulations have to be carried out. However, the problem of the comparison with the experimental results remains. Here we propose that the comparison would most easily be carried out in terms of the phase images and the displacement fields so calculated. The results obtained by a phase analysis of the simulations would then be compared with the experimental results. In this way, the disparities could easily be seen and quantified. It has also been argued that the local rotations are less affected by

the aberrations of the objective lens than the local lattice spacing.

In conclusion, the phase method is a rapid and quantitative way of producing maps of the deformation of the lattice with respect to a chosen reference. The method is general to all HREM images, and can be adapted to cope with differing noise levels and misalignments. However, the ease of interpretation of the results will depend on the experimental conditions and the materials studied. In most cases, it would be possible to test whether the lattice fringe positions no longer correspond to the atomic plane positions. The method can also be used in a more general way to produce a map of the local values of the basis vectors of the crystalline lattice and thus to describe, for example, the changes in the lattice across an interface.

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Appendix A

For calculations with digital images the coordinates $\mathbf{r}$ refer to the coordinates of pixels in the image. The reciprocal lattice vectors, $\mathbf{g}$, which appear in the equations, are in pixels^{-1} . This means that the coordinates of the reciprocal lattice vector in the numerical Fourier transform (with respect to the position $\mathbf{k} = 0$) must be divided by the size of the original image. If $\mathbf{g}$ is given by coordinates $\mathbf{g}_p = (g_{px}, g_{py})$ then:

$$g_x = g_{px}/N_x \quad \text{and} \quad g_y = g_{py}/N_y, \quad (\text{A.1})$$

where N_x and N_y are the horizontal and vertical dimensions of the image in pixels.

Appendix B

Consider a two-dimensional lattice defined in real space by the vectors, $\mathbf{a}_1$ and $\mathbf{a}_2$. The corresponding reciprocal lattice vectors, $\mathbf{g}_1$ and $\mathbf{g}_2$, are defined by the following relations:

$$\mathbf{g}_1 \cdot \mathbf{a}_1 = 1, \quad \mathbf{g}_1 \cdot \mathbf{a}_2 = 0, \quad (\text{B.1})$$

$$\mathbf{g}_2 \cdot \mathbf{a}_1 = 0, \quad \mathbf{g}_2 \cdot \mathbf{a}_2 = 1. \quad (\text{B.2})$$

Writing this in matrix form gives the following result:

$$\begin{pmatrix} g_{1x} & g_{1y} \\ g_{2x} & g_{2y} \end{pmatrix} \begin{pmatrix} a_{1x} & a_{2x} \\ a_{1y} & a_{2y} \end{pmatrix} = I, \quad (\text{B.3})$$

where I is the identity. Given the definitions, Eq. (36), for the matrices A and G , Eq. (B.3) becomes:

$$G^T A = I \quad (\text{B.4})$$

$$\text{or } (G^T)^{-1} = A: \begin{pmatrix} g_{1x} & g_{1y} \\ g_{2x} & g_{2y} \end{pmatrix}^{-1} = \begin{pmatrix} a_{1x} & a_{2x} \\ a_{1y} & a_{2y} \end{pmatrix}. \quad (\text{B.5})$$

Appendix C

It is sometimes useful to express the results in terms of the crystal lattice defined by the real space vectors, $\mathbf{a}$ and $\mathbf{b}$. In this case, the two reciprocal lattice vectors $\mathbf{g}_1$ and $\mathbf{g}_2$ can be written:

$$\mathbf{g}_1 = h_1 \mathbf{a}^* + k_1 \mathbf{b}^*, \quad (\text{C.1})$$

$$\mathbf{g}_2 = h_2 \mathbf{a}^* + k_2 \mathbf{b}^*, \quad (\text{C.2})$$

where h and k are the Miller indices for the reciprocal lattice vectors, $\mathbf{a}^*$ and $\mathbf{b}^*$. In matrix form, these equations become

$$\begin{pmatrix} g_{1x} & g_{1y} \\ g_{2x} & g_{2y} \end{pmatrix} = \begin{pmatrix} h_1 & k_1 \\ h_2 & k_2 \end{pmatrix} \begin{pmatrix} a_x^* & a_y^* \\ b_x^* & b_y^* \end{pmatrix}. \quad (\text{C.3})$$

Taking the inverse of both sides gives:

$$\begin{pmatrix} g_{1x} & g_{1y} \\ g_{2x} & g_{2y} \end{pmatrix}^{-1} = \begin{pmatrix} a_x^* & a_y^* \\ b_x^* & b_y^* \end{pmatrix}^{-1} \begin{pmatrix} h_1 & k_1 \\ h_2 & k_2 \end{pmatrix}^{-1}. \quad (\text{C.4})$$

Using on both sides of the equation, the relation given by Eq. (B.5) gives

$$\begin{pmatrix} a_{1x} & a_{2x} \\ a_{1y} & a_{2y} \end{pmatrix} = \frac{1}{h_1 k_2 - h_2 k_1} \begin{pmatrix} a_x & b_x \\ a_y & b_y \end{pmatrix} \times \begin{pmatrix} k_2 & -k_1 \\ -h_2 & h_1 \end{pmatrix}. \quad (\text{C.5})$$

Substituting this equation into Eq. (37) gives

$$\begin{pmatrix} u_x \\ u_y \end{pmatrix} = -\frac{1}{2\pi} \frac{1}{h_1 k_2 - h_2 k_1} \begin{pmatrix} a_x & b_x \\ a_y & b_y \end{pmatrix} \times \begin{pmatrix} k_2 & -k_1 \\ -h_2 & h_1 \end{pmatrix} \begin{pmatrix} P_{g1} \\ P_{g2} \end{pmatrix}, \quad (\text{C.6})$$

and therefore, in a similar way to Eq. (38), we find the result that

$$\mathbf{u}(\mathbf{r}) = -\frac{1}{2\pi} \left\{ \frac{k_2 P_{g1}(\mathbf{r}) - k_1 P_{g2}(\mathbf{r})}{h_1 k_2 - h_2 k_1} \mathbf{a} + \frac{h_1 P_{g2}(\mathbf{r}) - h_2 P_{g1}(\mathbf{r})}{h_1 k_2 - h_2 k_1} \mathbf{b} \right\}. \quad (\text{C.7})$$

Appendix D

The gradient of the phase is calculated by creating the complex image, $\exp\{iP_g(\mathbf{r})\}$, and then taking the gradient:

$$\frac{\partial}{\partial x} \exp\{iP_g(\mathbf{r})\} = i \frac{\partial P_g(\mathbf{r})}{\partial x} \exp\{iP_g(\mathbf{r})\}. \quad (\text{D.1})$$

Therefore,

$$\frac{\partial P_g(\mathbf{r})}{\partial x} = -i \exp\{-iP_g(\mathbf{r})\} \frac{\partial}{\partial x} \exp\{iP_g(\mathbf{r})\} \quad (\text{D.2})$$

$$= \Im m \left[\exp\{-iP_g(\mathbf{r})\} \frac{\partial}{\partial x} \exp\{iP_g(\mathbf{r})\} \right], \quad (\text{D.3})$$

where $\Im m$ is the imaginary part.

Appendix E

In general, the relation between the local lattice and the reference lattice is given by a *deformation*

matrix, $D(\mathbf{r})$, such that (see, e.g., Ref. [8]):

$$A(\mathbf{r}) = D(\mathbf{r})A. \quad (\text{E.1})$$

Taking the inverse of both sides and using the relationship $G^T = A^{-1}$ gives

$$G^T(\mathbf{r}) = G^T D^{-1}(\mathbf{r}). \quad (\text{E.2})$$

Rearranging this equation we have

$$D(\mathbf{r}) = [A G^T(\mathbf{r})]^{-1}. \quad (\text{E.3})$$

Substituting the expression for $G(\mathbf{r})$ in terms of $\Delta G(\mathbf{r})$, Eq. (43), gives the following result:

$$D(\mathbf{r}) = [I + A \Delta G^T(\mathbf{r})]^{-1}, \quad (\text{E.4})$$

such that the deformation matrix can be calculated from the phase images using Eq. (45).

This approach can be related to the strain matrices for small distortions. When the elements in the matrix ΔG are small, the right-hand side of Eq. (E.4) can be approximated to give

$$D(\mathbf{r}) \simeq I - A \Delta G^T(\mathbf{r}). \quad (\text{E.5})$$

For small distortions [8], the deformation matrix is given by

$$D(\mathbf{r}) \simeq I + e(\mathbf{r}) \quad (\text{E.6})$$

which gives

$$e(\mathbf{r}) = -A \Delta G^T(\mathbf{r}). \quad (\text{E.7})$$

Substituting the expression for $\Delta G(\mathbf{r})$ in terms of the phase images, Eq. (45), we can see that Eq. (E.7) is identical to Eq. (42).

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